School Worksheet

# Class 12 Mathematics, Chapter 6: Application of Derivatives

Maximum marks: **52** **20** questions

Name _____ Date _____ Score _____ / 52

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

| Q No. | Question | Marks |
|---|---|---|

## Level 1

8 marks

**1** MCQ — 1 mark

A cylindrical tank is being filled with sand at the rate of $314\ \text{m}^3/\text{h}$. If the radius of the tank is $10$ m, then the height of sand in the tank increases at the rate of: (Take $\pi = 3.14$)

**(A)** $1.1$ m/h **(B)** $1$ m/h **(C)** $\pi$ m/h **(D)** $\frac{\pi}{2}$ m/h

**2** MCQ — 1 mark

The function $f(x) = x^3 - 3x^2 + 3x - 100$ is:

**(A)** increasing on $\mathbb{R}$ **(B)** decreasing on $\mathbb{R}$

**(C)** increasing on $(-\infty, 1)$ only **(D)** decreasing on $(1, \infty)$

**3** MCQ — 1 mark

The interval in which $f(x) = x^2 - 4x + 6$ is strictly decreasing is:

**(A)** $(-\infty, 2)$ **(B)** $(2, \infty)$ **(C)** $(-\infty, \infty)$ **(D)** $(-2, 2)$

**4** MCQ — 1 mark

The maximum value of $\sin x \cos x$ is:

**(A)** $1$ **(B)** $\frac{1}{2}$ **(C)** $2$ **(D)** $\frac{1}{\sqrt{2}}$

**5** MCQ — 1 mark

The local minimum value of $f(x) = x + \frac{1}{x}$, $x > 0$, is:

**(A)** $1$ **(B)** $-2$ **(C)** $2$ **(D)** $0$

**6** Assertion-Reason — 1 mark

**Assertion (A):** $f(x) = \sin x$ is strictly increasing in $\left(0, \frac{\pi}{2}\right)$. **Reason (R):** If $f'(x) > 0$ for every $x$ in an open interval, then $f$ is strictly increasing in that interval.

**(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

**(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

**(C)** Assertion (A) is true, but Reason (R) is false.

**(D)** Assertion (A) is false, but Reason (R) is true.

| Q No. | Question | Marks |
|---|---|---|
| 7 | Short answer<br>The edge of a cube is increasing at the rate of $3$ cm/s. How fast is the volume increasing when the edge is $10$ cm long? | 2 marks |

## Level 2

19 marks

| | | |
|---|---|---|
| 8 | Short answer<br>Find the intervals in which $f(x) = 2x^3 - 3x^2 - 36x + 7$ is (a) strictly increasing (b) strictly decreasing. | 3 marks |
| 9 | Short answer<br>Air is pumped into a spherical balloon at the rate of $900\ \text{cm}^3/\text{s}$. Find the rate at which the radius is increasing when the radius is $15$ cm. | 2 marks |
| 10 | Short answer<br>Find the local maximum and local minimum values of $f(x) = x^3 - 6x^2 + 9x + 15$. | 3 marks |
| 11 | Short answer<br>Find the intervals in which the function $f(x) = \frac{\log x}{x}$ is strictly increasing or strictly decreasing. | 3 marks |
| 12 | Short answer<br>A ladder $5$ m long leans against a wall. The bottom of the ladder is pulled along the ground away from the wall at $2$ cm/s. How fast is its height on the wall decreasing when the foot of the ladder is $4$ m from the wall? | 3 marks |
| 13 | Short answer<br>Find the absolute maximum and absolute minimum values of $f(x) = \sin x + \cos x$ on $[0, \pi]$. | 3 marks |
| 14 | Short answer<br>Find two positive numbers whose sum is $15$ and the sum of whose squares is minimum. | 2 marks |

## Level 3

25 marks

| | | |
|---|---|---|
| 15 | Long answer<br>Show that of all rectangles inscribed in a given fixed circle, the square has the maximum area. | 5 marks |
| 16 | Long answer<br>Show that the height of the cylinder of maximum volume that can be inscribed in a sphere of radius $R$ is $\frac{2R}{\sqrt{3}}$. Also find the maximum volume. | 5 marks |
| 17 | Long answer<br>A window is in the form of a rectangle surmounted by a semicircular opening. The total perimeter of the window is $10$ m. Find the dimensions of the window so as to admit maximum light through the whole opening. | 5 marks |

| Q No. | Question | Marks |
|---|---|---|
| 18 | Short answer<br>Find the intervals in which $f(x) = \sin x + \cos x$, $0 \leq x \leq 2\pi$, is strictly increasing or strictly decreasing. | 3 marks |
| 19 | Short answer<br>A man $2$ m tall walks at a uniform speed of $5$ km/h away from a lamp post $6$ m high. Find the rate at which the length of his shadow increases. | 3 marks |
| 20 | Case study<br>**Case Study:** A technical company is designing a rectangular solar panel installation on a roof using $300$ metres of boundary material. The design includes a partition running parallel to one of the sides, dividing the area into two sections. Let the length of the side perpendicular to the partition be $x$ metres and the side parallel to the partition be $y$ metres.<br>**(i)** Write the equation for the total boundary material used (boundary and partition) in terms of $x$ and $y$. — 1 mark<br>**(ii)** Write the area of the solar panel as a function of $x$. — 1 mark<br>**(iii)(a)** Find the critical point of the area function and use the second derivative test to find the maximum area. — 2 marks<br>OR<br>**(iii)(b)** Using the first derivative test, find the maximum area the company can enclose with the $300$ m of boundary material. — 2 marks | 4 marks |

# Answer key

Chapter 6: Application of Derivatives, School Worksheet. Full step-wise solutions are in the solutions sheet.

## Level 1

| | |
|---|---|
| 1 | (B) $1$ m/h |
| 2 | (A) increasing on $\mathbb{R}$ |
| 3 | (A) $(-\infty, 2)$ |
| 4 | (B) $\frac{1}{2}$ |
| 5 | (C) $2$ |
| 6 | (A) Both A and R are true, and R is the correct explanation of A. |
| 7 | The volume is increasing at $900\ \text{cm}^3/\text{s}$. |

## Level 2

| | |
|---|---|
| 8 | (a) Strictly increasing on $(-\infty, -2) \cup (3, \infty)$; (b) strictly decreasing on $(-2, 3)$. |
| 9 | The radius is increasing at $\dfrac{1}{\pi}$ cm/s. |
| 10 | Local maximum value $= 19$ (at $x = 1$); local minimum value $= 15$ (at $x = 3$). |
| 11 | Strictly increasing on $(0, e)$; strictly decreasing on $(e, \infty)$. |
| 12 | The height on the wall is decreasing at $\dfrac{8}{3}$ cm/s. |
| 13 | Absolute maximum $= \sqrt{2}$ (at $x = \frac{\pi}{4}$); absolute minimum $= -1$ (at $x = \pi$). |
| 14 | The numbers are $\dfrac{15}{2}$ and $\dfrac{15}{2}$. |

## Level 3

| | |
|---|---|
| 15 | Area is maximum when $x = y = \sqrt{2}\,a$, i.e. the rectangle is a square. |
| 16 | Height $= \dfrac{2R}{\sqrt{3}}$; maximum volume $= \dfrac{4\pi R^3}{3\sqrt{3}}$ cubic units. |
| 17 | Width $= \dfrac{20}{\pi + 4}$ m, height of rectangle $= \dfrac{10}{\pi + 4}$ m (radius of semicircle $= \dfrac{10}{\pi + 4}$ m). |
| 18 | Strictly increasing on $\left[0, \frac{\pi}{4}\right) \cup \left(\frac{5\pi}{4}, 2\pi\right]$; strictly decreasing on $\left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$. |
| 19 | The shadow increases at $\dfrac{5}{2}$ km/h. |
| 20 | (i) $2x + 3y = 300$; (ii) $A(x) = 100x - \dfrac{2x^2}{3}$; (iii) maximum area $= 3750\ \text{m}^2$ at $x = 75$ m, $y = 50$ m. |